Kinetic Theory Of Ideal Gases (Assumptions):

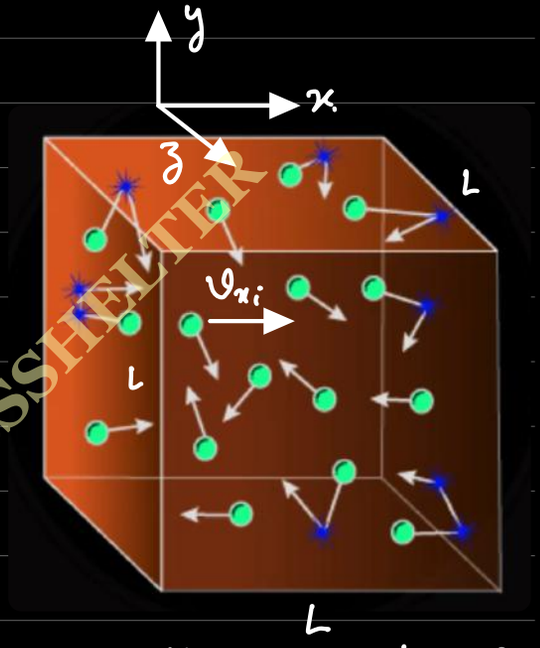
① Particles of ideal gas are always in Random motion making perfectly elastic collisions with vessel walls as well as each other.
② Volume of particles is negligible as compared to Volume occupied by the gas. (Particle volume is practically zero in comparison).
③ Force of interaction between the particles is absent.
④ Temperature of the Gas is proportional to its Kinetic Energy.

Derivation Of Ideal Gas Equation:

Consider a Cubical Box containing Gas particles.

Let:

m → mass of each Particle,
M → molar mass of Gas
n → Total number of moles of gas
N → Total numbe of particles of gas
$\vec{v}_i$ → Velocity of ith Particle.



Now obviously: $nM = Nm$ [∵ Total mass of gas written in different ways].

Now consider the ith particle moving with x component of velocity as v_{x_i}. When it collides with right wall, its change in momentum

$$\Delta p_{x_i} = 2mv_{x_i} \quad (\because \text{elastic collision}).$$

Now such collisions will keep happening periodically with time period:

$$\Delta t_i = 2L/v_{x_i} \quad (\because \text{particle covers } 2L \text{ distance in a round trip})$$

Now average force exerted by this particle on right wall:

$$F_{x_i} = \Delta p_{x_i}/\Delta t_i = (mv_{x_i}^2/L) \quad ①$$

Now total force on right wall:

$$F_x = \Sigma F_{x_i} = \frac{m}{L}\Sigma v_{x_i}^2 \quad ②$$

Now ∵ there is no directional preference in motion of gas particles

$$\therefore \quad \Sigma v_{x_i}^2 = \Sigma v_{y_i}^2 = \Sigma v_{z_i}^2 = \Sigma (v_{x_i}^2 + v_{y_i}^2 + v_{z_i}^2)/3 = \frac{1}{3}\Sigma v_i^2 \quad ③$$

Root Mean Square Velocity: For any sample of particles, RMS velocity

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is defined as $v_{rms} = \sqrt{\frac{\Sigma v_i^2}{N}} \Rightarrow \Sigma v_i^2 = N v_{rms}^2$ (4)

Using (4) in (3) and the putting Σv_i^2 in (2):

$$F_x = \frac{1}{3}\frac{m}{L} N v_{rms}^2$$

Now Pressure on right wall: $F_x/A = F_x/L^2 = \frac{mN}{3L^3} v_{rms}^2 = P$ (5)

Now mN/L^3 is Total mass/Total Volume i.e Density (ρ).

$$\Rightarrow \quad P = \frac{1}{3}\rho v_{rms}^2 \Rightarrow \boxed{v_{rms} = \sqrt{\frac{3P}{\rho}}}$$

Now Rearranging (5) and using $V = L^3$:

$$PV = \frac{nM}{3} v_{rms}^2 \quad (6) \qquad (\because Nm = nM)$$

Also KE of the system: $KE = \frac{1}{2}\Sigma m v_i^2 = \frac{1}{2} m \Sigma v_i^2$

$$\Rightarrow \quad KE = \frac{1}{2} mN v_{rms}^2 = \frac{1}{2} nM v_{rms}^2 \quad (7)$$

Now $\frac{1}{2} M v_{rms}^2$ represents molar kinetic energy which is proportional to temperature. $\Rightarrow \frac{1}{2} M v_{rms}^2 = CT$ where C is some proportionality constant.

Using this in (6):

$PV = n \times \frac{2}{3} CT$, now we can put $\frac{2}{3} C$ as another constant $R \Rightarrow$:

$\boxed{PV = nRT}$ (8) $\Rightarrow \quad P = \frac{n}{V} RT = \frac{nM}{V} \cdot \frac{1}{M} RT = \rho RT/M$

$\Rightarrow \quad \boxed{P = \rho RT/M}$ (9) Alternate form (I) of ideal gas equation.

We define another constant known as Boltzmann constant k as

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$k = R/N_A$ $\quad N_A \rightarrow$ Avogadro Number, $R \rightarrow$ Universal Gas Constant.

Rearranging (8): $P = \left(\frac{N}{N_A}\right)\cdot\frac{1}{V}RT = \left(\frac{N}{V}\right)\left(\frac{R}{N_A}\right)T$

$\Rightarrow$ $\boxed{P = \eta kT}$ (10) Here η is particle density (particles/unit Volume)

RMS Velocity in terms of T and M: Using $P = \rho RT/M$ in rms velocity formula, we can write:

$$\boxed{v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}}} \quad (11)$$

Total KE Of Point Particle Gas: Using (11) in (7):

$$U = KE = \frac{1}{2}nMv_{rms}^2 = \frac{3}{2}nRT = \frac{3}{2}PV$$

$\Rightarrow$ $\boxed{U = \frac{3}{2}nRT = \frac{3}{2}PV}$ (12)

Note: $R = 8.314\ J/mol\ K \approx 2\ Cal/mol\ K = 0.0821\ lit\ atm/mol\ K.$
$k = 1.38 \times 10^{-23}\ J/K$ per particle.

Ideal Gas Laws: $PV = nRT \Rightarrow$

At constant Temperature $V \propto 1/P$ (Boyle's Law)
At constant Pressure $V \propto T$ (Charles Law)
At constant Volume $P \propto T$ (Gay-Lussac's Law).

[These laws were discovered before rigorous development of ideal Gas Theory, but all of them follow from ideal Gas equation).

Example: The best Vaccuum obtained so far is 10^{-10} mm of Hg. Find the particle density corresponding to this at 27°C.

Solution: $P = \eta kT \Rightarrow \eta = P/kT$

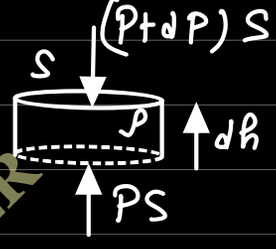
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$$\Rightarrow \eta = \frac{10^{-10}}{760} \times \frac{1.01 \times 10^5}{1.38 \times 10^{-23} \times 300} \text{ particles/m}^3$$

$$\Rightarrow \eta = 3.21 \times 10^{12} \text{ particles/m}^3. \quad \text{Ans.}$$

BAROMETRIC FORMULA: Assuming constant temperature and acceleration due to gravity, derive an expression for pressure as a function of height h from earth's surface assuming P_0 at the surface.

Solution: Consider FBD of a small column of air, $\rho S g dh$



By force balance: $(P+dP)S + \rho S dh g = PS$

$$\Rightarrow dP = -\rho g dh \quad \text{Also } P = \rho RT/M \Rightarrow \rho = PM/RT$$

$$\Rightarrow dP = -\frac{PMg}{RT} dh \Rightarrow \frac{dP}{P} = -\frac{Mg}{RT} dh$$

$$\Rightarrow \int_{P_0}^{P} \frac{dP}{P} = -\frac{Mg}{RT} \int_0^h dh \Rightarrow \boxed{P = P_0 e^{-Mgh/RT}}$$

IRODOV EXAMPLE:

2.1. A vessel of volume $V = 30$ l contains ideal gas at the temperature 0 °C. After a portion of the gas has been let out, the pressure in the vessel decreased by $\Delta p = 0.78$ atm (the temperature remaining constant). Find the mass of the released gas. The gas density under the normal conditions $\rho = 1.3$ g/l.

Solution: $pV = \frac{m}{M} RT \Rightarrow \Delta p V = \Delta m \frac{RT}{M}$

$$\Rightarrow \Delta m = \frac{MV}{RT} \Delta p, \text{ Now } \rho = \frac{P_0 M}{R T_0} \quad (P_0 = 1 \text{ atm}, T_0 = 273 \text{ K})$$

$$\Rightarrow M = \rho R T_0 / P_0 \Rightarrow \Delta m = \frac{\rho R T_0}{P_0} \times \frac{V}{RT} \Delta p = \rho V \frac{T_0}{T} \cdot \frac{\Delta p}{P_0} = \rho V \frac{\Delta p}{P_0}$$

$$\Rightarrow \Delta m = 1.3 \text{ g/l} \times 30 \text{ l} \times 1 \times \frac{0.78}{1} = 30 \text{ g}$$

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PARTIAL PRESSURES (GAS MIXTURE): If we have a mixture of two (or more) gases, then net pressure on vessel walls is due to collisions from both type of species. Let p_1 and p_2 be their individual pressure contributions then:

$$p_1 V = n_1 RT \;,\; p_2 = n_2 RT \;,\; p = p_1 + p_2 = (n_1 + n_2) RT/V$$

$$\Rightarrow \frac{p_1}{p} = \frac{n_1}{n_1 + n_2} \quad \text{similarly} \quad \frac{p_2}{p} = \left(\frac{n_2}{n_1 + n_2}\right)$$

Effective molar mass of a mixture:

$$P = p_1 + p_2 = \rho_1 RT/M_1 + \rho_2 RT/M_2 \quad (1)$$

$$\Rightarrow P = RT\left[\frac{\rho_1}{M_1} + \frac{\rho_2}{M_2}\right], \text{ Also: } \rho = \left(\frac{n_1 M_1 + n_2 M_2}{V}\right), \rho_1 = \frac{n_1 M_1}{V}, \rho_2 = \frac{n_2 M_2}{V}$$

$$\Rightarrow \rho_1 = \left(\frac{n_1 M_1}{n_1 M_1 + n_2 M_2}\right)\rho \quad \text{also} \quad \rho_2 = \left(\frac{n_2 M_2}{n_1 M_1 + n_2 M_2}\right)\rho,$$

Putting this in (1):

$$P = RT\left[\frac{n_1 + n_2}{n_1 M_1 + n_2 M_2}\right]\rho = \frac{\rho RT}{M_{eff}} \quad \text{where } M_{eff} = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2}$$

$$\Rightarrow \boxed{P = \rho RT/M_{eff}} \quad \text{where} \quad \boxed{M_{eff} = (n_1 M_1 + n_2 M_2)/(n_1 + n_2)}$$

→ The formula can be extended to a number of gases as:

$$\boxed{M_{eff} = \Sigma n_i M_i / \Sigma n_i}$$

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EXAMPLE (IRODOV):

2.3. A vessel of volume $V = 20$ l contains a mixture of hydrogen and helium at a temperature $t = 20\ ^\circ\text{C}$ and pressure $p = 2.0$ atm. The mass of the mixture is equal to $m = 5.0$ g. Find the ratio of the mass of hydrogen to that of helium in the given mixture.

Solution: $PV = \left(\frac{m_1}{M_1} + \frac{m_2}{M_2}\right)RT$ ① $\quad m = m_1 + m_2$ ②

From ① & ②, $m_1 = \frac{M_1(M_2PV - mRT)}{(M_2 - M_1)RT}$, $m_2 = \frac{M_2(M_1PV - mRT)}{(M_2 - M_1)RT}$

$\Rightarrow \frac{m_1}{m_2} = \frac{M_1}{M_2}\left(\frac{M_2PV - mRT}{mRT - M_1PV}\right)$, Dividing numerator & Den by M_1M_2PV

$\Rightarrow \frac{m_1}{m_2} = \frac{(1 - a/M_2)}{(a/M_1 - 1)} = 0.5$ (Here $a = (mRT/PV)$)

EXAMPLE (IRODOV):

2.7. A vessel of volume V is evacuated by means of a piston air pump. One piston stroke captures the volume ΔV. How many strokes are needed to reduce the pressure in the vessel η times? The process is assumed to be isothermal, and the gas ideal.

Let P_n be the pressure after n^{th} stroke then

$\because$ process is isothermal $\Rightarrow$:

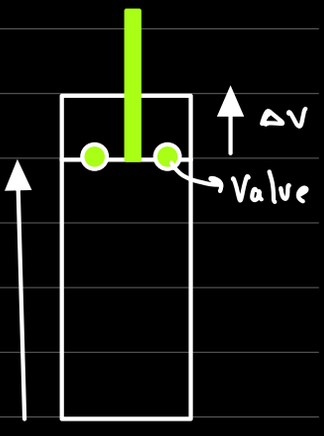
$$P_nV = P_{n+1}(V + \Delta V)$$

$\Rightarrow P_{n+1} = P_n\left(\frac{V}{V+\Delta V}\right) \Rightarrow$ Pressure decreases in G.P.

$\Rightarrow P_n = P_0\left(\frac{V}{V+\Delta V}\right)^n$ Now $\frac{P_n}{P_0} = \frac{1}{\eta}$

$\Rightarrow \ln\eta = i\ln\left(\frac{V+\Delta V}{V}\right)$, Here i is number of required strokes.

$\Rightarrow i = \ln\eta / \ln\left(\frac{V+\Delta V}{V}\right) = \ln\eta / \ln(1 + \Delta V/V)$ Ans.



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IRODOV EXAMPLE:

2.8. Find the pressure of air in a vessel being evacuated as a function of evacuation time t. The vessel volume is V, the initial pressure is p_0. The process is assumed to be isothermal, and the evacuation rate equal to C and independent of pressure.

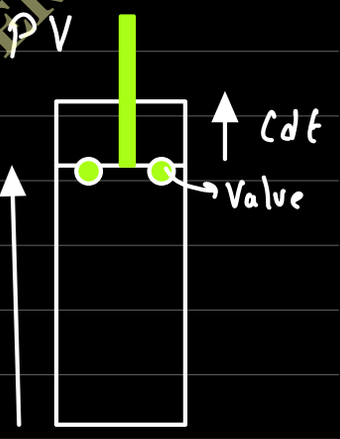
Note. The evacuation rate is the gas volume being evacuated per unit time, with that volume being measured under the gas pressure attained by that moment.

Solution: This is an extension of previous problem where infinitesimal air volume is being captured but the process is happening continuous. Let pressure at any time be P and after a time dt be $P+dP$ $\Rightarrow$ $(P+dP)(V+Cdt) = PV$

$\Rightarrow$ $PCdt + VdP = 0$

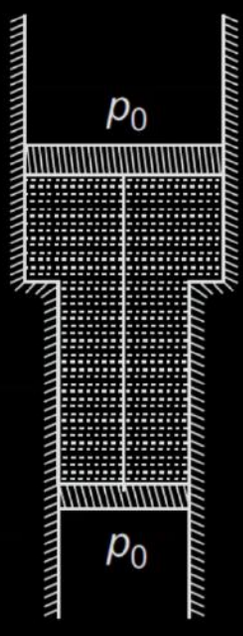
$\Rightarrow$ $\frac{dP}{P} = -\frac{C}{V}dt$ $\Rightarrow$ $\ln\frac{P}{P_0} = -\frac{Ct}{V}$

$\Rightarrow$ $P = P_0 e^{-Ct/V}$ Ans.



IRODOV 2.10

A smooth vertical tube having two different sections is open from both ends and equipped with two pistons of different areas (Fig. 2.1). Each piston slides within a respective tube section. One mole of ideal gas is enclosed between the pistons tied with a non-stretchable thread. The cross-sectional area of the upper piston is $\Delta S = 10\,\text{cm}^2$ greater than that of the lower one. The combined mass of the two pistons is equal to $m = 5.0\,\text{kg}$. The outside air pressure is $p_0 = 1.0$ atm. By how many kelvins must the gas between the pistons be heated to shift the pistons through $l = 5.0\,\text{cm}$?



Solution: Consider the Combined FBD of the pistons and the string.

For equilibrium $(P-P_0)\times\Delta S = mg$ (1) $\Rightarrow$ Process must be isobaric

$\because$ $P = (mg/\Delta S) + P_0$ Now when pistons move upward by a distance l, $\Delta V = l\Delta S$ (Volume will increase on heating).

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Now $PV = nRT \Rightarrow \Delta T = \frac{P\Delta V}{nR} = \left(\frac{mg}{\Delta s} + P_0\right)\cdot\frac{l\Delta s}{nR}$

$n = 1 \Rightarrow$

$\Delta T = (mg + P_0\Delta s)\, l/R$ Ans.

IRODOV EXAMPLE: (2.12)

Find the minimum attainable pressure of ideal gas in the process $T = T_0 + \alpha V^2$, where T_0 and α are positive constants, and V is the volume of one mole of gas. Draw the approximate p vs V plot of this process.

Solution: $T = T_0 + \alpha V^2$, $PV = RT \Rightarrow P = \frac{R(T_0 + \alpha V^2)}{V}$

For minimum Pressure, $dP/dV = 0 \Rightarrow (-T_0/V^2) + \alpha = 0$

$\Rightarrow \alpha V^2 = T_0$ & $V = \sqrt{\frac{T_0}{\alpha}} \Rightarrow P_{min} = R(2T_0)\sqrt{\frac{\alpha}{T_0}} = 2R\sqrt{\alpha T_0}$ Ans.

IRODOV 2.14:

Suppose the pressure p and the density ρ of air are related as $p/\rho^n = \text{const}$ regardless of height (n is a constant here). Find the corresponding temperature gradient. $(dT/dy = ?)$

Solution: $dP = -\rho g\, dy$ ① Now $P = C\rho^n \Rightarrow dP = nC\rho^{n-1}d\rho$

Using dP in ①

$\Rightarrow nC\rho^{n-2}d\rho = -g\,dy$ ②

Now $P = \rho RT/M \Rightarrow C\rho^{n-1} = \frac{RT}{M} \Rightarrow (n-1)C\rho^{n-2} = \frac{R}{M}dT$ ③

③/② $\Rightarrow \frac{-R}{Mg}\frac{dT}{dy} = \left(\frac{n-1}{n}\right) \Rightarrow \frac{dT}{dy} = \frac{-Mg}{R}\left(\frac{n-1}{n}\right)$ Ans.

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Solution: $P/\rho^n = C$ ① $dP = -\rho g\, dy$ ② $P = \rho RT/M$ ③

From ① & ③, $C\rho^{n-1} = RT/M$ ④

From ① $dP = Cn\rho^{n-1} d\rho$ ⑤ using in ②:

$$Cn\rho^{n-2} d\rho = -g\, dy \quad ⑥$$

From ④, $(n-1)C\rho^{n-2} d\rho = \frac{R}{M} dT$ ⑦

From ⑦ & ⑥ $\therefore -\frac{R}{Mg}\frac{dT}{dy} = \left(\frac{n-1}{n}\right) \Rightarrow \frac{dT}{dy} = -\frac{Mg}{R}\left(\frac{n-1}{n}\right)$

REAL GASES: In real gases, the volume of molecules is not literally zero like we assumed for ideal gas, nor is intraction force between particles zero. It turns out that at very low separations this force is repulsion, while above a certain separation it becomes attraction in nature. Due to these differences, the equation of state for real gas does not strictly obeys the ideal gas equation, however at low pressures and high temperatures, the behavior becomes close to the ideal gas behavior.

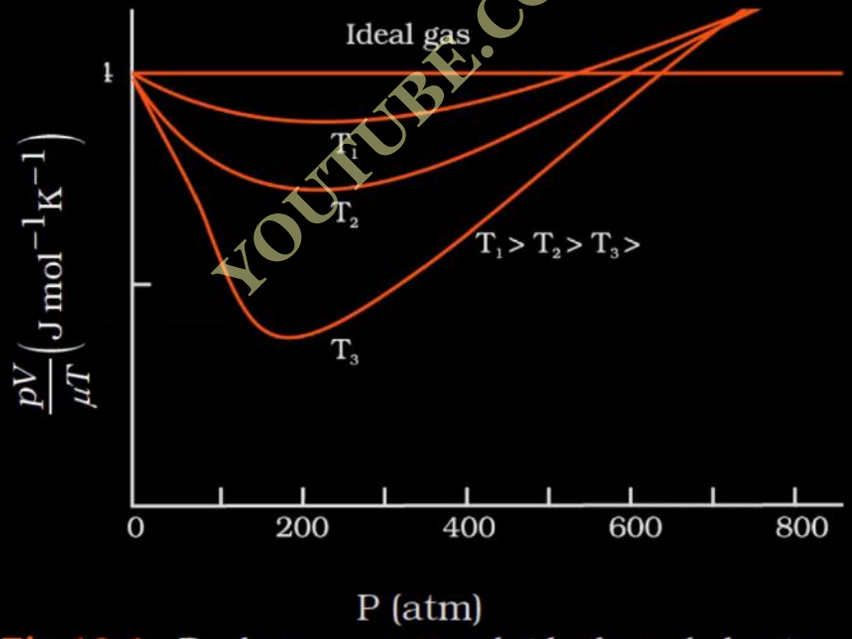


Fig.13.1 *Real gases approach ideal gas behaviour at low pressures and high temperatures.*

← Variation of PV/nT for ideal Gas & Real Gas.

For same Volume & Temperature, Real Gas exerts smaller pressure at low pressures due to inward attraction of molecules, Whereas at high pressures, repulsive forces dominate and at same volume & T, Pressure becomes higher.

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NCERT:

▶ ***Example 13.1*** The density of water is 1000 kg m^{-3}. The density of water vapour at 100 °C and 1 atm pressure is 0.6 kg m^{-3}. The volume of a molecule multiplied by the total number gives ,what is called, molecular volume. Estimate the ratio (or fraction) of the molecular volume to the total volume occupied by the water vapour under the above conditions of temperature and pressure.

Sol: Ignoring the spacing between water molecules in liquid form, total molecular volume can be estimated to be same as volume of liquid. Since total mass remains same, ratio of volumes is in inverse ratio of densities

$\therefore$ Volume of Vapour is $\frac{1000}{0.6} = \frac{1}{6\times10^{-4}}$ times that of liquid

$\therefore$ molecular volume is 6×10^{-4} times the volume of the vapour. (or Vapour volume is 1.67×10^{3} times the molecular volume).

NCERT:

EXAMPLE 13.2: Estimate volume of a water molecule using data in previous example. Also estimate its radius (Assuming spherical). and spacing between two molecules.

Solution: Mass of water molecule $m = 18g / 6\times10^{23}$

Volume $= m/\rho = \frac{18\times10^{-3}}{6\times10^{23}} \times \frac{1}{1000}\ m^3 = 3\times10^{-29}\ m^3$

Now: $V = 4/3\ \pi R^3 \Rightarrow R = (3V/4\pi)^{1/3} \approx 2\text{Å}$

$\therefore$ Spacing between molecules $= 2R = 4\text{Å}$.

NCERT:

EXAMPLE 13.3: What is average spacing between water molecules upon vaporisation (Same data as example 13.1).

Sol: Volume increases by a factor 1.67×10^{3} $\Rightarrow$ spacing increases by a factor of $(1.67\times10^{3})^{1/3}$ ($\because V \propto R^3$).

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$\Rightarrow$ Spacing increases by a factor of ≈ 10.

$\Rightarrow$ new spacing $= 4\text{Å} \times 10 \approx 40\text{Å}$ Ans.

NCERT: Dalton's Law Of Partial Pressures:

> **Example 13.4** A vessel contains two non-reactive gases : neon (monatomic) and oxygen (diatomic). The ratio of their partial pressures is 3:2. Estimate the ratio of (i) number of molecules and (ii) mass density of neon and oxygen in the vessel. Atomic mass of Ne = 20.2 u, molecular mass of O_2 = 32.0 u.

Solution: According to Dalton's law of Partial Pressures, Partial Pressures are directly proportional to number of moles.

$\therefore n_{Ne} : n_{O_2} = 3:2$ Ans.

Now mass density of Ne $= n_{Ne} \times M_{Ne} / V$

" " " O_2 $= n_{O_2} \times M_{O_2} / V$

$$\Rightarrow \rho_{Ne}/\rho_{O_2} = \frac{n_{Ne}}{n_{O_2}} \times \frac{M_{Ne}}{M_{O_2}} = \frac{3}{2} \times \frac{20.2}{32} = 0.947$$

Maxwell Distribution Of Gas Particle Velocities: In any sample of Gas, the particles are in continuous random motion with various velocities. A statistical analysis gives us the probability of finding a number of particles between certain speeds and such a probability function which gives us the distribution of speeds in a sample is called Maxwell Boltzmann Distribution for Gas particle velocities.

We shall not do the complete derivation of the distribution function however we shall do some analysis to get insights in to the basis of Derivation of velocity distribution function.

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Let total number of particles in the system be N and let total Energy of the system be E.
Further let N_1 particle be at energy E_1, N_2 particles be at Energy E_2 and so on. Then according to macroscopic constraints :

$$\Sigma N_i = N \quad (1)$$
$$\Sigma N_i E_i = E \quad (2)$$

(The particles are considered identical but distinguishable).
For instance consider we have 100 particles and total 1000 Joule Energy to be distributed), One type of microstates that could satisfy this macroscopic constraint would be to give 100 Joule Energy each to any 10 Particles and nothing to remaining 90 particles). Now subject to macroscopic constraints, there will be certain number of permutations to attain any particular type of microstates. Based on logic of combinatonics, it can be easily inferred that certain types of microstates will have very large number of possible permutations. (Mole factorials! are involved here). Due to random exchanges, system keeps on randomly migrating from one microstate to another. However since certain types of microstates are exponentially larger than other states, upon random observation, most probably the system will be found in one of these states and hence the probability distribution of finding particles speeds in various ranges.
A rigorous statistical analysis using logic based on permutations combinations of very large samples and Calculus gives us the following formulae regarding distributions of particle velocities in a gas sample at a general temperature T.

$$dN(v) = N\left(\frac{m}{2\pi kT}\right)^{3/2} \times e^{\frac{-mv^2}{2kT}} \times 4\pi v^2 dv \quad (1)$$

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Here: $N \rightarrow$ Total number of particles,
$dN \rightarrow$ Number of particles with speed between v & $v+dv$
$m \rightarrow$ particle mass
$k \rightarrow$ Boltzmann constant
$T \rightarrow$ Temperature
$v \rightarrow$ Speed of the particles

From equation ① it can be readily seen that number of particles moving with very high velocity will be negligible because for very high velocities, exponential term will go to zero. Similarly number of particles moving with very low velocities will also be very small because at very low velocities, $4\pi v^2$ term will be very small.

PROBABILITY DENSITY FUNCTION: From equation ① above, it can be seen that fraction of particles between velocities v & $v+dv$ is $\frac{dN}{N}$ this can also be called probability of finding a particle between speed v & $v+dv$. Obviously larger the interval dv, larger will be the number of particles in the interval. ∴ we can also define probability per unit velocity interval i.e $(dN/N)/dv$, this is known as probability density function. Mathematically:

$$f(v) = \frac{1}{N}\frac{dN}{dv} = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{mv^2}{2kT}} \times 4\pi v^2 \quad ②$$

Then probability of finding a particle between v_1 & v_2 is simply:

$$P(v_1 \rightarrow v_2) = \frac{\Delta N}{N} = \int_{v_1}^{v_2} f(v)\,dv \quad ③$$

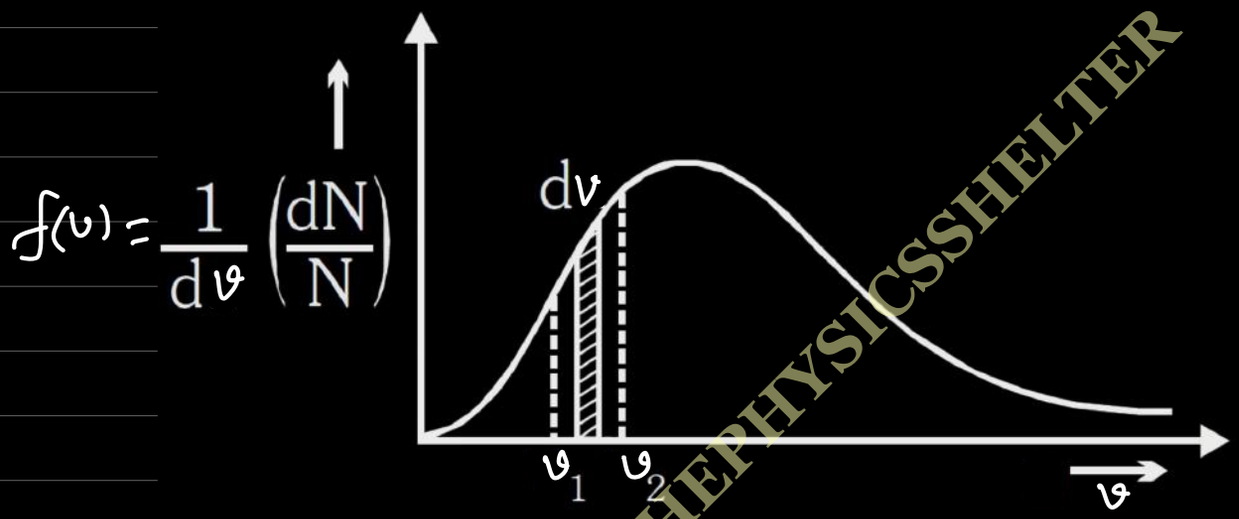
Which is simply area under the $f(v)$ vs v graph.
Also we can find total number of particles in this interval as
$\Delta N = N \times P(v_1 \rightarrow v_2)$

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$$\text{or} \quad \Delta N = N \int_{v_1}^{v_2} f(v)\, dv$$

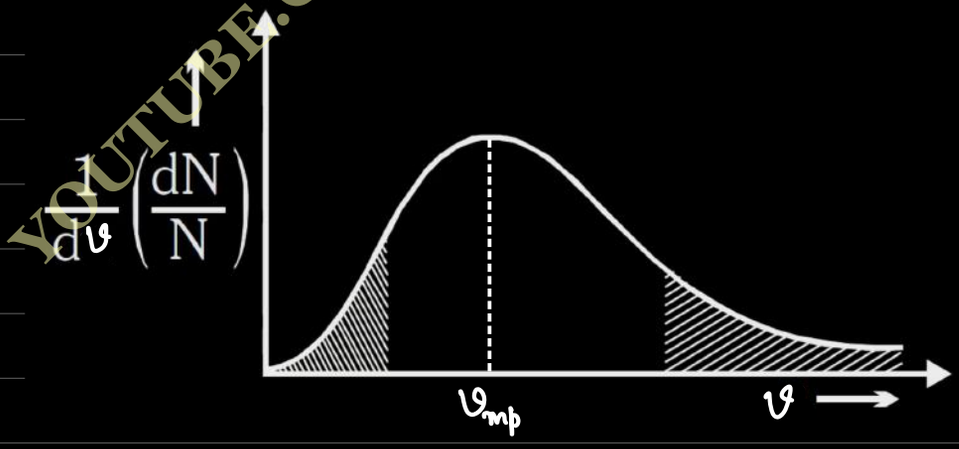
It can also be readily infered from equation ③ that total area under $f(v)$ graph from 0 to ∞ velocity must be unity because between $v = 0$ to $v = \infty$, $\Delta N = N$ (All particles have a velocity between 0 and ∞).

Graph Of Maxwell Boltzmann Distribution:



SHADED AREA → Probability of finding a particle between v & $v + dv$

Properties Of Maxwell Boltzmann Distribution Graph:



(I) Relatively fewer particles have very high or very low velocity.

(II) The graph peaks at a point where probability density is constant the corresponding velocity is known as most probable velocity (v_{mp}). This can be calculated using Calculus. i.e for most probable speed:

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$\frac{df(v)}{dv} = 0$. Upon maximizing $f(v)$ we get:

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2kT}{m}}$$

(III) Average Velocity : Average velocity of the species is defined as $v_{avg} = (\Sigma N_i v_i)/N$, for a continuous distribution,

$$v_{avg} = \frac{1}{N}\int_0^\infty v\,dN = \frac{1}{N}\int_0^\infty v_x f(v) x N dv = \int_0^\infty v f(v) dv$$

Upon calculation, we get:

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8kT}{\pi m}}$$

(IV) RMS Velocity (Root mean square):

Root mean square of a system of particles is defined as

$$v_{rms} = \sqrt{\frac{\Sigma N_i v_i^2}{N}}$$, for a continuous distribution,

it can be written as $v_{rms} = \sqrt{\frac{1}{N}\int_0^\infty v^2\,dN}$.

$$\Rightarrow v_{rms} = \sqrt{\int_0^\infty f(v) v^2}$$

Upon calculation, this comes out to be:

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

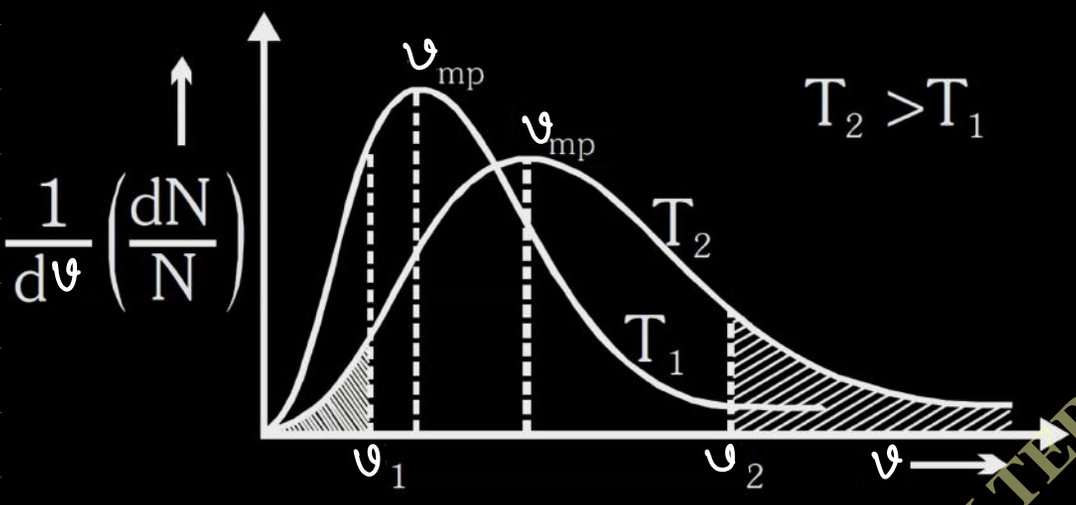
Note that this is the same value that we earlier calculated by applying kinetic theory of gases to particles in a cubical Box.

We can see that $v_{rms} > v_{avg} > v_{mp}$

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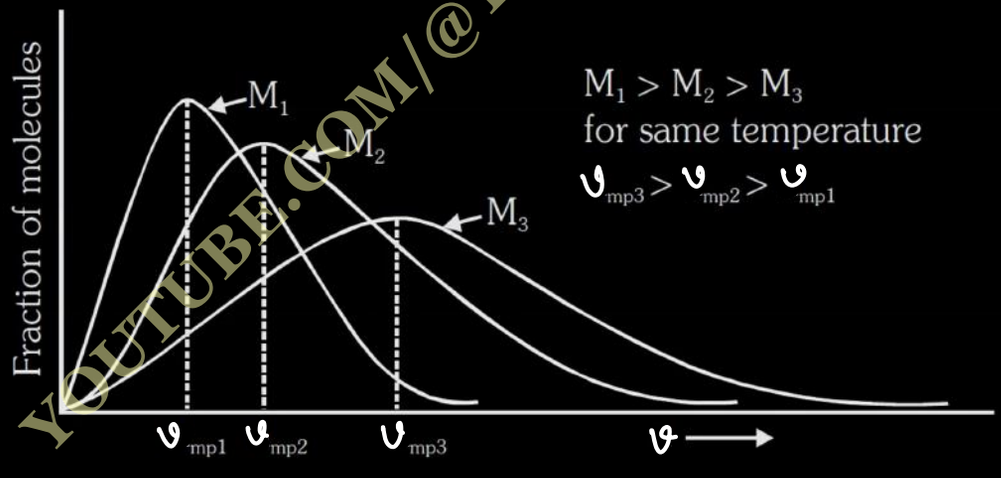
Also $v_{rms} : v_{avg} : v_{mp} = 1 : 0.92 : 0.816$.

(IV) Change in plot of $f(v)$ with temperature:



As temperature increases, fraction of particles moving at higher speeds increases and at lower speeds decreases. Also, fraction of particles near most probable velocities decreases.

(V) Effect of Molar Mass:



Note that $f(v)$ depends on the ratio (M/T) thus effect of molar mass is opposite of the temperature, if molar mass and temperature are increased in same ratio then distribution remains same.

Note: Most probable Kinetic Energy Does not correspond to most probable velocity, this might sound counter intuitive but remember that this is a continuous distribution and not a discrete one. (Discrete intuition fails hua). This can be readily seen as follows:

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$$dN(v) = N\left(\frac{m}{2\pi kT}\right)^{3/2} \times e^{\frac{-mv^2}{2kT}} \times 4\pi v^2 dv$$

Now $E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E}{m}} \Rightarrow dv = \sqrt{\frac{2}{m}} \cdot \frac{1}{2\sqrt{E}} dE$

$$dN(E) = 2\pi\left(\frac{1}{\pi kT}\right)^{3/2} \sqrt{E}\, e^{-E/kT} dE,$$ Here the form of the function on RHS has changed altogether, so it should not be surprising that, the maxima would occur for some E which does not correspond to most probable speed. (This is because same velocity intervals do not correspond to same Kinetic Energy intervals. Infact same velocity interval corresponds to a larger Kinetic Energy interval for higher speeds $\because dE = mv\,dv$.

Other Forms Of Maxwell Distribution Function:

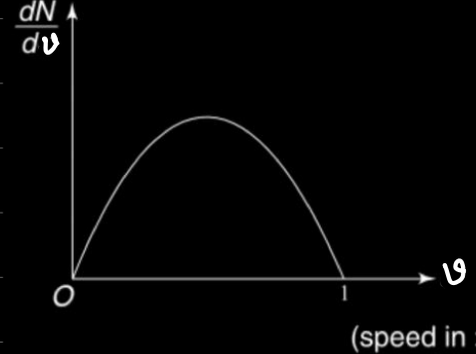
(I) Component-Wise: $dN(v_n) = N\left(\frac{m}{2\pi kT}\right)^{1/2} e^{-mv_n^2/kT} dv_n$

(II) Reduced Form:

$$dN(u) = N\frac{4}{\sqrt{\pi}} e^{-u^2} du,$$

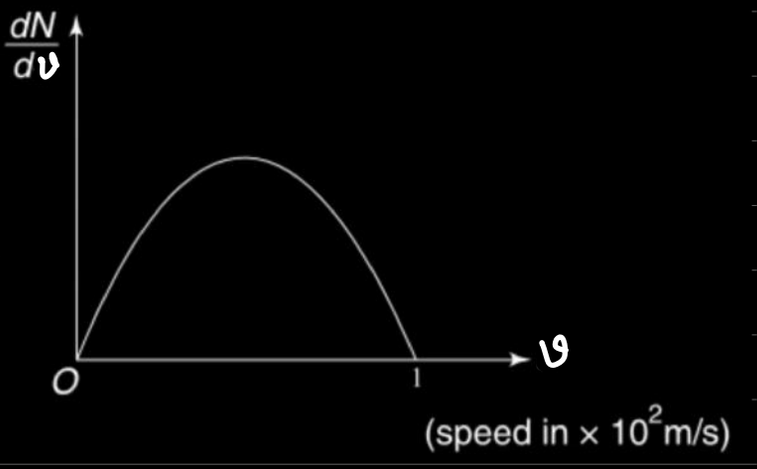
here $u = v/v_{mp}$ (Ratio of speed to most probable speed).

Example: The speed distribution of molecules in sample of gas varies quadratically with speed and total molecules is N_0.
(a) Calculate RMS speed of molecules
(b) Calculate Translational KE of molecules if mass of sample is 10g.



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Solution: Let $\frac{dN}{dv} = v(v-1) \times C$



where C is appropriate constant and v is in the given units on graph.
Now area under the graph must be N_0.

$$\Rightarrow \int_0^1 C(v^2 - v)\,dv = N_0$$

$$\Rightarrow C\left(\frac{1}{3} - \frac{1}{2}\right) = N_0 \Rightarrow C = -6N_0$$

$$\Rightarrow dN = -6N_0(v^2 - v)\,dv$$

Now by definition of rms: $v_{rms} = \sqrt{\frac{1}{N_0}\int v^2\,dN}$

$$\Rightarrow v_{rms} = \sqrt{\int_0^1 -6(v^4 - v^3)\,dv} \Rightarrow v_{rms} = \sqrt{-6\left(\frac{1}{5} - \frac{1}{4}\right)} = \sqrt{\frac{3}{10}}$$

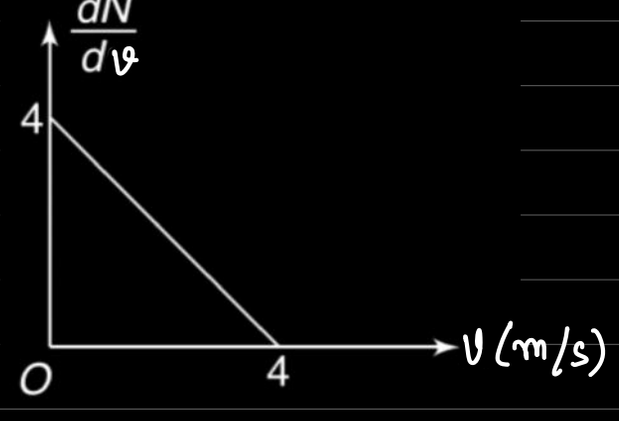
$$\Rightarrow v_{rms} = \sqrt{\frac{3}{10}} \times 10^2\ m s^{-1}$$

Now $KE = \frac{1}{2}\sum m_i v_i^2 = \frac{1}{2}\sum m_i \left(\sqrt{\frac{\sum m_i v_i^2}{\sum m_i}}\right)^2$

$$= \frac{1}{2} m v_{rms}^2 = \frac{1}{2} \times 10 \times 10^{-3} \times \frac{3}{10} \times 10^4 = 15\ \text{Joule}$$

$\therefore$ KE = 15 Joules Ans.

EXAMPLE: Find RMS and average speed for molecular sample with shown distribution. Also find number of particles.



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Sol: $\frac{dN}{dv} = -v + 4 \Rightarrow N = \left[-\frac{v^2}{2} + 4v\right]_0^4 = 8$ Ans

Now $v_{rms} = \sqrt{\frac{1}{N}\int v^2 dN} = \sqrt{\frac{1}{8}\int_0^4 (-v^3 + 4v^2)dv}$

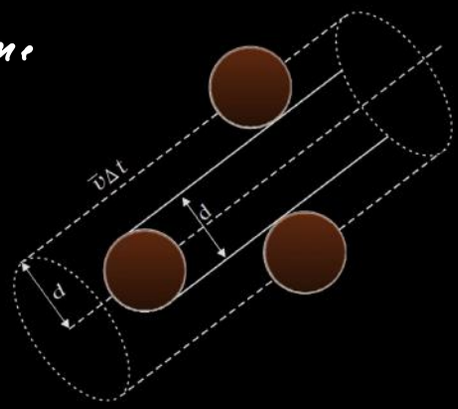
$\Rightarrow v_{rms} = \sqrt{\frac{1}{8}\left[-\frac{v^4}{4} + \frac{4v^3}{3}\right]_0^4} = \sqrt{\frac{8}{3}}\ ms^{-1}$

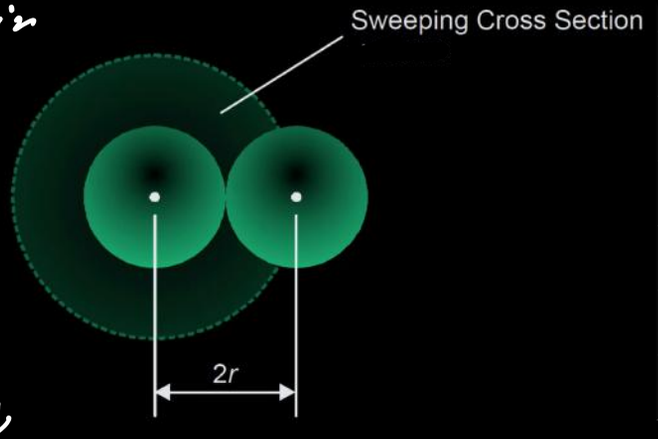
$v_{avg} = \frac{1}{N}\int v\,dN = \frac{1}{8}\int_0^4 -v^2 + 4v\,dv = \frac{1}{8}\left[-\frac{v^3}{3} + 2v^2\right]_0^4$

$\Rightarrow v_{avg} = \frac{4}{3}\ ms^{-1}$

COLLISION FREQUENCY: The number of collisions that a gas particle makes with other particles per unit time is called collision frequency. We shall try to estimate the order of collision frequency with a simplistic model and then state the exact formula which can be derived using a more rigorous model.

Let number density of gas particles per unit volume be η and let d be the diameter of each gas particle, let $\bar{v}$ be the average speed of gas particles, then for purpose of counting collisions, effective cross section radius of this particle is $2r$ (see figure), i.e. any particle whose centre comes within a distance of $2r$ from the centre of this particle gets swept by it. Thus in a time Δt,





→ Swept Volume = $\pi D^2 \times \bar{v}\Delta t$

→ Swept Particles = $\eta \pi D^2 \times \bar{v}\Delta t$

$\Rightarrow$ Number of particles swept per unit time,

$z = \eta \pi D^2 \bar{v}$

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However in this analysis we have assumed other particles to be stationary, although they are also moving, thus actual collision frequency will be higher than our estimate. It turns out that for actual frequency, we need to multiply our expression by $\sqrt{2}$.

Thus Collision frequency:

$$\nu = \sqrt{2}\,\eta\,\pi D^2 \bar{v}$$

Mean relaxation time (τ): Average time elapsed between two successive collisions is called mean relaxation time. Mathematically it should be reciprocal of collision frequency. i.e $\tau = 1/\nu$

$$\tau = 1/\sqrt{2}\,\eta\,\pi D^2 \bar{v}$$

→ Remember to use average speed here and not RMS speed i.e $\sqrt{\frac{8RT}{\pi M}}$.

Mean Free Path: Average distance travelled by gas particles between two successive collisions is called mean free path (λ). Mathematically it should be product of average speed & mean relaxation time.

i.e : $\lambda = \bar{v} \times \tau$

$$\Rightarrow \quad \lambda = 1/\sqrt{2}\,\eta\,\pi D^2$$

→ Note that mean free path is independent of temperature even though mean relaxation time & $\bar{v}$ are dependent on temperature.

EXAMPLE: Estimate mean free path of air molecules at STP assuming diameter of particles to be 2 Å. Also find τ assuming $\bar{v} = 485\ ms^{-1}$

Solution: 1 mole occupied 22.4 lit $\Rightarrow \eta = 6.023 \times 10^{23} / 22.4 \times 10^{-3}$

$$\Rightarrow \lambda = \frac{1}{\sqrt{2}\,\eta\,\pi D^2} = \frac{22.4 \times 10^{-3}}{\sqrt{2} \times 6.023 \times 10^{23} \times \pi \times (2 \times 10^{-10})^2} = 2.09 \times 10^{-7}\ m \approx 1000\ d$$

Now $\lambda = \bar{v}\tau \Rightarrow \tau = \lambda / \bar{v} = 2.09 \times 10^{-7} / 485 = 4.3 \times 10^{-10}$ sec.

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NUMBER OF COLLISIONS/UNIT AREA: Using Maxwell's Distribution and Kinetic Theory, it can be shown that number of collisions per unit area per unit time is given by:

$$\gamma_A = \frac{1}{4} n \bar{v}$$

If there is a very small opening in Gas tank, (of the order of mean free path or smaller), the gas leak through such an opening is called **EFFUSION**, and from above formula it is evident that is should be proportional to $\sqrt{T/M}$ for same particle density $\because \bar{v} = \sqrt{8RT/\pi M}$.

DIFFUSION: When gases mix in to each other due to different concentration, (on hole of order much greater than mean free path), the phenomenon is called diffusion. Just like effusion, rate of diffusion is also inversely proportional to square root of molar mass. (This is known as Graham's law of diffusion.).

EXAMPLE (200 More Puzzling 101 Q.)

P101* There is a *very small* hole in an otherwise totally enclosed heated furnace. Outside the furnace, the air temperature is 0 °C and the air pressure is 100 kPa. The air inside the furnace is kept at a constant temperature of 57 °C by the controlled heating system, and after a sufficiently long time its pressure also becomes stationary. Estimate the magnitude of this stationary pressure.

Solution: In the steady state, the rate of effusion of air from inside to outside must be exactly equal to rate of effusion from outside to inside.

$$\Rightarrow n_i \bar{v}_i = n_o \bar{v}_o \qquad \text{Now } PV = nRT \Rightarrow \frac{n}{V} = P/RT$$

$$\text{Now } n = \frac{n}{V} \times N_A \text{ and } \bar{v} \propto \sqrt{\frac{RT}{M}} \Rightarrow n\bar{v} \propto \frac{P}{RT} \times \sqrt{\frac{RT}{M}} \propto \frac{P}{\sqrt{MT}}$$

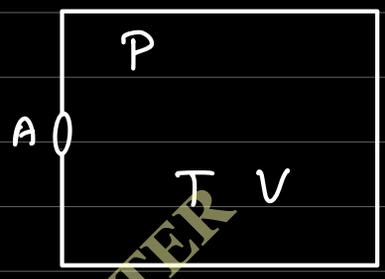
$$\Rightarrow \frac{P_i}{\sqrt{T_i}} = \frac{P_o}{\sqrt{T_o}} \Rightarrow P_i = P_o \sqrt{\frac{T_i}{T_o}} = 10^5 \sqrt{\frac{330}{273}} \approx 1.1 \times 10^5 \text{ Pa}.$$

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Example (NCERT Exemplar):

A box of 1.00 m^3 is filled with nitrogen at 1.50 atm at 300 K. The box has a hole of an area 0.010 mm^2. How much time is required for the pressure to reduce by 0.10 atm, if the pressure outside is 1 atm.

Solution: $P = \eta kT \Rightarrow \frac{dP}{dt} = kT \frac{d\eta}{dt}$ ①



Now in a time dt, particles effusing out = $dN_i = (\eta_i \bar{v}/4\, dt)\, A$

Similarly particles effusing inside = $(\eta_0 \bar{v}/4) A\, dt$

$\Rightarrow \left(\frac{dN}{dt}\right)_{net} = -(\eta_i - \eta_0)\frac{\bar{v}}{4} A \Rightarrow \frac{d\eta}{dt} = \left(\frac{dN}{dt}\right) \times \frac{1}{V}$

$\Rightarrow \frac{d\eta}{dt} = -(\eta_i - \eta_0)\frac{\bar{v}}{4}\frac{A}{V}$, Putting in ①:

$$\frac{dP}{dt} = -(\eta_i - \eta_0)\frac{\bar{v}}{4} \cdot \frac{A}{V} \times kT = -\left(\frac{P - P_0}{kT}\right)\frac{\bar{v}}{4}\frac{A}{V} \times kT \quad (\because \eta = P/kT)$$

$$\Rightarrow \left(\frac{dP}{P - P_0}\right) = \frac{-A\bar{v}}{4V} dt \Rightarrow \ln\left(\frac{P_f - P_0}{P_i - P_0}\right) = \frac{-A\bar{v}}{4VkT} t$$

$$\Rightarrow t = \frac{4V}{A\bar{v}} \ln\left(\frac{P_i - P_0}{P_f - P_0}\right), \quad \bar{v} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8 \times 8.314 \times 300}{\pi \times 28 \times 10^{-3}}} = 476\ ms^{-1}$$

$$\Rightarrow t = \frac{4 \times 1}{0.01 \times 10^{-6} \times 476} \times \ln\left(\frac{5}{4}\right) = 2.4 \times 10^5 \text{ Seconds} \approx 2.8 \text{ days}.$$

→ (Answer in Exemplar is Incorrect).

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Van Der Waal's Equation Of State:

While developing the ideal gas theory, we had assumed force of interaction between particles to be zero and also assumed Gas molecules to be point size particles, while infact the gas particles have some force of interaction and do infact occupy a finite volume no matter how so small.
Van Der Waal try to come up with a considerably better model for Gases taking in to account these factors.

Accounting For Interaction Forces:

Consider one mole of an ideal gas, we can write for this:

$$PV = RT \Rightarrow P = \frac{RT}{V}$$

However for a real gas, due to attraction forces from other atoms, it is expected that the pressure will be smaller. Now it turns out that

1. Force of attraction on any particle is proportional to number of particles surrounding it (or particle density),
2. Number of particles per unit area near the wall is also proportional to particle density. ∴

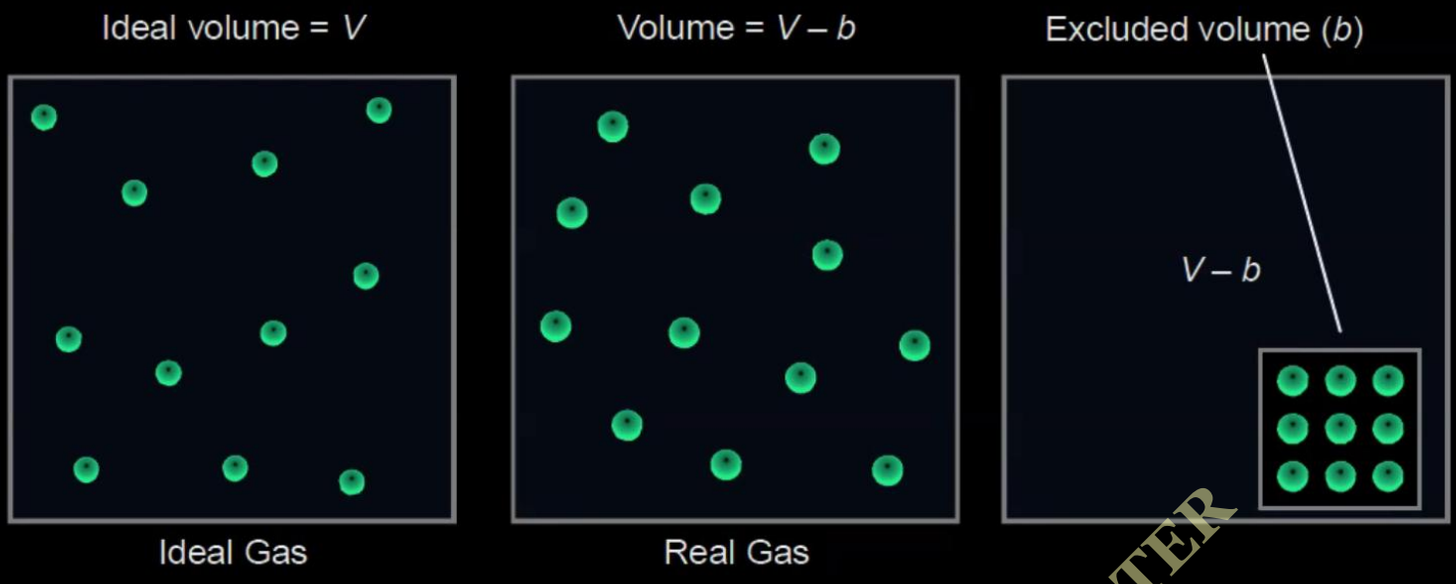
Considering these two factors the reduction in pressure can be considered proportional to the square of particle density, or inversely proportional to molar specific volume (Volume per mole, V_m)
Thus considering this factor our equation should become:

$$P = \frac{RT}{V_m} - \frac{a}{V_m^2} \quad \text{or} \quad \left(P + \frac{a}{V_m^2}\right) V_m = RT \quad (1)$$

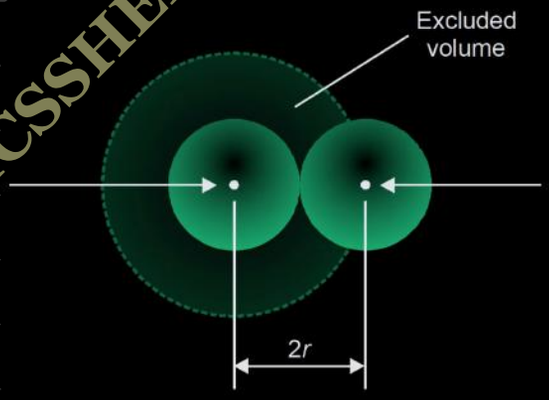
Here a is some proportionality constant depending on strength of particle interactions.

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Accounting For Excluded Volume:



Considering gas particles to be like solid balls, it can be readily seen that centres of any two balls cannot come any closer than $2r$. Thus volume excluded for any pair of particles is $\frac{4}{3}\pi \times (2r)^3$, $\therefore$ per particle, excluded volume becomes: $\frac{1}{2} \times \frac{4}{3}\pi (2r)^3$



$\Rightarrow$ excluded volume per particle $= \frac{16\pi r^3}{3} = 4 V_{molecule}$.

Now Ideal Gas equation:

$$PV = RT \Rightarrow V = RT/P$$

By accounting for excluded Volume due to finite volumes of molecules, ideal gas equation should be modified as:

$$V - b = RT/P \Rightarrow P(V-b) = RT \quad (2)$$

Now combining the effects responsible for equations (1) & (2), we get:

$$\boxed{\left(P + \frac{a}{V_m^2}\right)\left(V_m - b\right) = RT} \quad (3)$$

Here V_M = Volume/mol, b = Excluded Volume/mol = 4 x Volume of 1 mol molecules.

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For n mols of gas, equation (3) can be suitably modified as:

$$\left(P+\frac{a}{(V/n)^2}\right)\left(\frac{V}{n}-b\right)=RT \qquad (\because V_m = V/n)$$

multiplying by n throughout:

$$\boxed{\left(P+\frac{an^2}{V}\right)(V-nb)=nRT} \qquad (4)$$

Potential Energy Of A Van Der Waal Gas: Recall that change in potential energy is negative of work done by conservative forces. When a Van Der Waal gas is expanded, it has to do so against internal attraction force. Now imagine a sphere containing 1 mol of ideal gas expanding against internal interaction pressure of a/V_m^2, let the volume expand by dV_m.

$$\Rightarrow \quad dPE = -dW_{int} = -\left(\frac{a}{V_m^2}\,dV_m \times \cos(180^\circ)\right) = \frac{a}{V_m^2}\,dV_m$$

$$\Rightarrow \quad PE = \frac{-a}{V_m} + C \quad \text{where } C \text{ is is suitable constant, depending}$$

on reference for zero potential energy. For convenience we choose $C=0$

$$\Rightarrow \quad \boxed{PE = \frac{-a}{V_m}}$$

Total Energy Of Van-Der Waal Gas: Apart from potential energy, a Van Der Waal gas will also have a kinetic energy which is just a function of temperature. Thus total Energy of Van Der Waal gas is:

$$U = f(T) - \frac{a}{V_m} \qquad \text{(for one mol)}.$$ Now when $V_m \to \infty$, we know that Van-Der Waal gas should behave like ideal gas ($\because$ attraction force will become negligible, and so also fraction of excluded volume)

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$\therefore \quad f(T) = U_{ideal.} \Rightarrow U_{vw} = U_{ideal} - \frac{a}{V_m}$

$\Rightarrow \boxed{U_{vw} = C_v T - \frac{a}{V_m}}$ Here C_v refers to specific heat at constant volume about which we will learn in a later section.

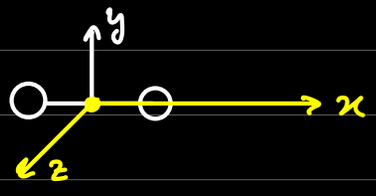
Degrees of Freedom: For a gas species with atomicity N, each atom can posess kinetic Energy by virtue of its velocity components along x, y and z directions. Thus we say that each atom has three degrees of freedom and the molecules has a total of 3N degrees of freedom. (On kinetic Energy of molecule can be described using 3N Quadratic terms).

Translational Degrees of Freedom: The number of terms required to describe kinetic energy of a molecule if it is in pure translation (i.e neither rotating nor vibrating) is known as translational degrees of freedom. If a molecule does not rotate or vibrate, it can only posess KE by virtue of its translation in x, y or z directions. Thus every molecule has 3 translational Degrees of freedom.

Rotational Degrees of Freedom: Minimum number of terms required to describe Kinetic Energy of a molecule in its CM frame if it is not vibrating is known as rotational degrees of freedom.
Rotational degrees of various types of molecules are described below:

① Monoatomic Gas : 0 rotational degrees (∵ moment of inertia is zero)

② Diatomic Gas:



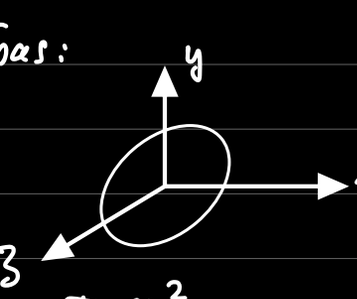
$I_x = 0, \; I_y \& I_z \neq 0$

$\Rightarrow KE = \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2$ (Principal axes)

$\Rightarrow$ Rotational Degrees of freedom = 2.

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③ Non linear polyatomic Gas:



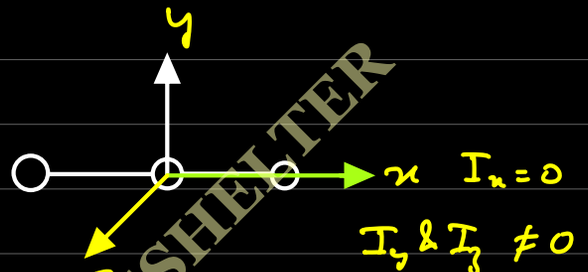
$I_x \neq 0,\ I_y \neq 0,\ I_z \neq 0$

$$KE = \frac{1}{2} I_x \omega_x^2 + \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2$$

(Recall $KE = \frac{1}{2} \vec{L} \cdot \vec{\omega}$ and $\vec{L} = I_x \omega_x \hat{i} + I_y \omega_y \hat{j} + I_z \omega_z \hat{k}$ if x, y & z are principal axes)

$\Rightarrow$ Rotational Degrees of freedom = 3

④ Polyatomic linear Molecule:



As clear from figure this is just like diatomic molecule, from rotation point of view. ∴ Rotational Degrees = 2.

VIBRATIONAL DEGREES OF FREEDOM: A molecule can also possess energy without translation of its CM or change of orientation by just vibrations of its bonds (or flexing of bonds). The number of modes in which a molecule can possess vibrational energy is called vibrational degrees of freedom. No of vibrational degrees is calculated by subtracting other degrees of freedom from total degrees of freedom. Thus if atomicity is N,

$$f_{vib} = 3N - f_{trans} - f_{rot}$$, For instance for water, $f_{vib} = 3 \times 3 - 3 - 3 = 3$ Thus there are three vibrational degrees (various ways in which molecule can possess energy by bond flexing / vibrations). Molecules possess vibrational Energy only at sufficiently high temperatures and it is governed by laws of Quantum Mechanics. In most problems of JEE Physics, vibrational degrees of freedom are ignored unless implied or explicitly stated.

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Based on above discussion we can say that excluding vibrational degrees of freedom,

For monoatomic Gas, $f = 3$ $(3+0)$
For diatomic Gas, $f = 5$ $(3+2)$
For polyatomic Gas, $f = 6$ $(3+3)$

Law of Equipartition of Energy: According to law of equipartition, for any dynamic system in thermal equilibrium, total energy of the system is equally divided among degrees of freedom.

Recall that for point particles gas, we had shown that kinetic energy of n moles at T temperature is given as:

$$U = \frac{3}{2} nRT = \frac{3}{2} PV, \text{ Thus energy per degree of freedom}$$

per particles is given as: $u = \left(\frac{\frac{3}{2} nRT}{3 \times n \times N_A} \right) = \frac{1}{2} kT$ $(\because f = 3)$.

Thus kinetic energy per particle per degree of freedom is $\frac{1}{2} kT$
Now using equipartition, we can write KE of n moles of a gas with f degrees of freedom as:

$$U = n \times N_A \times f \times \frac{1}{2} kT = n \frac{f}{2} RT$$

$$\Rightarrow \boxed{U = \frac{nfRT}{2}}$$

First Law Of Thermodynamics: If dQ is the amount of Energy (Heat) supplied to a system, dU is its change in internal energy and dW is the work done by the system then:

$$dQ = dU + dW$$

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(This is based on conservation of Energy). Since this equation is true for infinitesimals, it is also true for finites, i.e.:

$\Delta Q = \Delta U + \Delta W$ or Net heat supplied to any system is equal to sum of change in its internal energy and work done by the system.

Molar Specific Heat of a Gas: Heat required per mol per unit rise in temperature of a gas is called molar specific heat. Mathematically, if dQ is the heat required to raise temperature of n moles of a gas by an amount dT then specific heat is given as:

$$\boxed{C = \frac{1}{n}\frac{dQ}{dT}} \Rightarrow dQ = nC\,dT \Rightarrow \Delta Q = nC\Delta T \quad [\text{If Constant } C]$$

(C: molar)

Molar specific heat in general will depend on the process involved (on how much work is extracted from Gas while supplying the heat, more the amount of work extracted, higher will be specific heat).

C_v (Specific heat at Constant Volume): Here as the name suggests, a gas is heated at constant volume. (Piston is locked).

Since the displacement of piston is zero, Gas is not doing any work ($\because dW = \vec{F}\cdot d\vec{s}$ here even though $\vec{F} \neq 0$ but $d\vec{s} = 0$).



Now $dQ = dU + dW = dU \quad (\because dW = 0)$

Now $U = n\frac{f}{2}RT \Rightarrow dU = n\frac{f}{2}R\,dT$

$\Rightarrow dQ = n\frac{f}{2}R\,dT \Rightarrow C_v = \frac{1}{n}\frac{dQ}{dT} = \frac{f}{2}R$

Thus $\boxed{C_v = \frac{f}{2}R}$

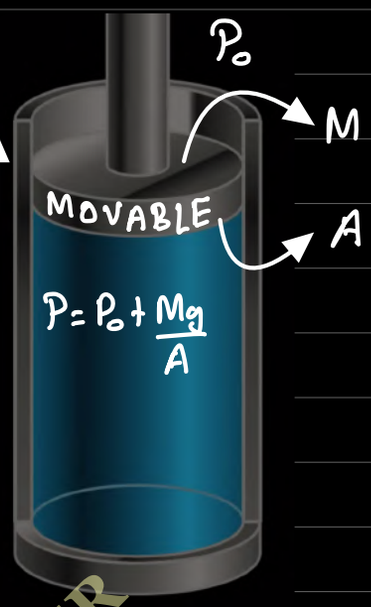
Now $\because U = n\frac{f}{2}RT \Rightarrow \boxed{U = nC_vT}$ (C_v is just an alias of $\frac{f}{2}R$).

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C_p (Specific Heat at Constant Pressure):

Here as the name suggests, piston is free to move maintaining constant pressure $(P_0 + Mg/A)$

Now $dQ = dU + dW$, $dW = PA\,dx = P\,dV$



Now:

$$\frac{1}{n}\frac{dQ}{dT} = \frac{1}{n}\frac{dU}{dT} + \frac{1}{n}\frac{dW}{dT} \quad (1)$$

$$U = n\frac{f}{2}RT \Rightarrow \frac{1}{n}\frac{dU}{dT} = \frac{f}{2}R \quad (2)$$

Also $dW = P\,dV \Rightarrow \frac{dW}{dT} = P\frac{dV}{dT}$ (3)

$$PV = nRT \Rightarrow P\,dV = nR\,dT = dW \Rightarrow \frac{1}{n}\frac{dW}{dT} = R \quad (4)$$

Using (4) & (2) in (1), $\frac{1}{n}\frac{dQ}{dT} = \frac{f}{2}R + R$

$$\Rightarrow \boxed{C_p = \left(1 + \frac{f}{2}\right)R}$$

Gamma of a Gas (Heat Capacity Ratio / Adiabatic Index / Laplace Coeff):

Ratio of Specific Heats at constant pressure to that at constant Volume is known as Gamma of a Gas (γ): Mathematically:

$$\boxed{\gamma = C_p/C_v} \quad (1)$$

Now $C_p = \left(1 + \frac{f}{2}\right)R$ & $C_v = \frac{f}{2}R \Rightarrow \boxed{C_p - C_v = R}$ (2)

From (1) & (2):

$$\boxed{C_v = \frac{R}{\gamma - 1}} \qquad \boxed{C_p = \frac{\gamma R}{\gamma - 1}}$$

Also $C_p/C_v = \frac{1 + f/2}{f/2} = 1 + \frac{2}{f} \Rightarrow \boxed{\gamma = 1 + \frac{2}{f}}$

$\Rightarrow \gamma_{mono} = 1 + \frac{2}{3} = \frac{5}{3}$, $\gamma_{di} = 1 + \frac{2}{5} = \frac{7}{5} = 1.4$ $\gamma_{poly} = 1 + \frac{2}{6} = \frac{4}{3}$

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Effective Degrees of Freedom of a mixture: Recall that $U = n\frac{f}{2}RT$

now say we have a mixture of two gases with degrees of freedom f_1 and f_2 and n_1 & n_2 number of moles respectively, then:

$$U = n_1\frac{f_1}{2}RT + n_2\frac{f_2}{2}RT = (n_1+n_2)\frac{f_{eff}}{2}RT \quad (1)$$

Here f_{eff} is effective degrees of freedom for the mixture. Now by rearranging equation (1):

$$\boxed{f_{eff} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}} \quad (2)$$

Effective Gamma Of a Mixture: Recall that $\gamma = 1 + 2/f \Rightarrow f = \frac{2}{\gamma - 1}$

$$\Rightarrow \gamma_{eff} = 1 + 2/f_{eff} \Rightarrow \frac{2}{\gamma_{eff}-1} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

$$\Rightarrow \frac{2}{\gamma_{eff}-1} = \frac{n_1\,(2/(\gamma_1-1)) + n_2\,(2/(\gamma_2-1))}{(n_1+n_2)}$$

$$\Rightarrow \boxed{\frac{1}{\gamma_{eff}-1} = \left(\frac{1}{n_1+n_2}\right)\left[\frac{n_1}{\gamma_1-1} + \frac{n_2}{\gamma_2-1}\right]}$$

Note that above formula is very cumbersome and complicated and should in general be avoided. To find γ_{eff}, first find f_{eff} using equation (2) above and then find $\gamma_{eff} = 1 + 2/f_{eff}$.

→ $\gamma_{eff} \neq (n_1\gamma_1 + n_2\gamma_2)/(n_1+n_2)$ ← BLUNDER ALERT

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EXAMPLE (IRODOV): A certain Van Der Waal Gas is at temperature T, what should be the pressure p for density to be ρ. Molar mass is M and constants a and b are known.

Solution: $\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT$ ① $V_m \rightarrow$ molar volume.

$V_m = M/\rho$ ② Rearranging ①: $p = \left(\frac{RT}{V_m - b}\right) - \frac{a}{V_m^2}$

$\Rightarrow p = \frac{RT}{(M/\rho - b)} - \frac{a\rho^2}{M^2} \Rightarrow p = \frac{\rho RT}{(M - \rho b)} - \frac{a\rho^2}{M^2}$ Ans.

IRODOV EXAMPLE: (2.22)

One mole of nitrogen is contained in a vessel of volume $V = 1.00$ l.

Find:

the temperature of the nitrogen at which the pressure can be calculated from an ideal gas law with an error $\eta = 10\%$ (as compared with the pressure calculated from the Van der Waals equation of state);

Solution: By ideal Gas equation: $P = RT/V$

By Van Der Waal equation: $P' = \left(\frac{RT}{V - b}\right) - \frac{a}{V^2}$

Now $\frac{P - P'}{P'} = \eta \Rightarrow \frac{P}{P'} = (\eta + 1) \Rightarrow \frac{RT/V}{RT/(V-b) - a/V^2} = (\eta + 1)$

Rearranging and solving for temperature:

$T = a(V-b)(\eta+1)/RV(\eta V + b)$ Ans.

EXAMPLE (IRODOV 2.57):

Find the work performed by one mole of a Van der Waals gas during its isothermal expansion from the volume V_1 to V_2 at a temperature T.

Solution: $dW = PdV = \left[\left(\frac{RT}{V-b}\right) - \frac{a}{V^2}\right] dV \Rightarrow W = \int_{V_1}^{V_2} P\,dV$

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$$\Rightarrow \quad \Delta W = RT \ln\left(\frac{V_2 - b}{V_1 - b}\right) + a\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$$

Example (Irodov 2.58):

One mole of oxygen is expanded from a volume $V_1 = 1.00$ l to $V_2 = 5.0$ l at a constant temperature $T = 280$ K. Calculate:
(a) the increment of the internal energy of the gas:
(b) the amount of the absorbed heat.
The gas is assumed to be a Van der Waals gas.

Solution: Recall for Van Der Waal Gas, $U = n\left(C_v T - a/V_m\right)$

$\Rightarrow \Delta U = n\left(C_v \Delta T - a\left[\frac{1}{V_2} - \frac{1}{V_1}\right]\right) = a\left(\frac{1}{V_1} - \frac{1}{V_2}\right)$ Ans.

$\because \Delta T = 0$

(b) $\Delta Q = \Delta U + \Delta W$, we can use ΔW from previous example.

$\Rightarrow \Delta U = a\left(\frac{1}{V_1} - \frac{1}{V_2}\right) + RT \ln\left(\frac{V_2 - b}{V_1 - b}\right) + a\left[\frac{1}{V_2} - \frac{1}{V_1}\right]$

$\Rightarrow \Delta Q = RT \ln\left(\frac{V_2 - b}{V_1 - b}\right)$ Ans.

Example (Irodov 2.59): For a Van Der Waal gas find the difference of molar heat capacities $(C_p - C_v)$ as a function of T and V.

Solution: $C = \left(\frac{dQ}{dT}\right)$ for 1 mol $= \left(\frac{dU}{dT}\right)_{1\,mol} + \left(\frac{dW}{dT}\right)_{1\,mol}$

Now $U = \frac{f}{2}RT - \frac{a}{V_m}$, At constant Volume:

$\frac{dW}{dT} = 0$ ($\because dV = 0 \therefore PdV = 0$), Also $\because V_m = \text{Const}^{tt} \Rightarrow \frac{dU}{dT} = \frac{f}{2}R$

$\Rightarrow C_v = \frac{f}{2}R$ ①

Now at Constant Pressure: $\frac{dU}{dT} = \frac{f}{2}R + \frac{a}{V_m^2} \cdot \left(\frac{\partial V_m}{\partial T}\right)_P$ ②

$\frac{dW}{dT} = P\frac{dV_m}{dT} \Rightarrow C_p = \frac{dU}{dT} + \frac{dW}{dT} = \frac{f}{2}R + \left(\frac{\partial V_m}{\partial T}\right)_P \left(P + \frac{a}{V_m^2}\right)$ ③

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From (3) & (1), $C_p - C_v = \left(\frac{\partial V_m}{\partial T}\right)_p \left(P + a/V_m^2\right)$ (4)

Now $P = \left(\frac{RT}{V_m - b}\right) - \frac{a}{V_m^2}$, Differentiating at Constant Pressure:

$$0 = R\left[\frac{1}{V_m - b} + T\left[\frac{-1}{(V_m - b)^2} \times \frac{dV_m}{dT}\right]\right] + \frac{2a}{V_m^3}\frac{dV_m}{dT}$$

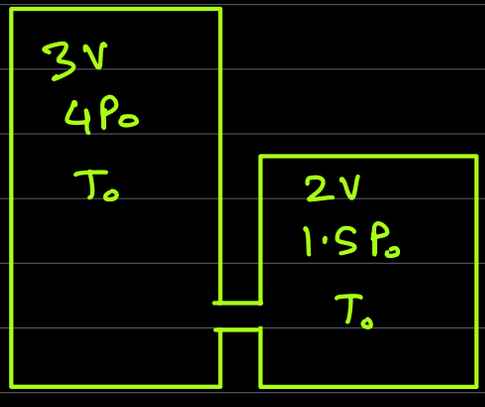
$$\Rightarrow \frac{dV_m}{dT}\left[\frac{2a}{V_m^3} - \frac{RT}{(V_m - b)^2}\right] + \frac{R}{V - b} = 0 \quad (5)$$

Using (5) in (4): $C_p - C_v = \left[\frac{R/(V-b)}{(RT/(V_m-b)^2) - 2a/V_m^3}\right]\frac{RT}{(V_m - b)}$

$$\Rightarrow C_p - C_v = \frac{R}{1 - \frac{2a(V_m - b)^2}{RT\,V_m^3}} \quad \text{Ans.}$$

EXAMPLE:

A vessel of volume 3V contains gas at pressure $4P_0$ and another vessel of volume 2V contains the same gas at pressure $1.5P_0$. Both vessels have same temperature. When both vessels are connected by a tube of negligible volume, the equilibrium pressure is $nP_0/2$, Where n is an integer. Find the value of n (Assume no heat exchange with surroundings)



Solution: Let final pressure be P.

By conserving moles:

$$\frac{4P_0 \times 3V}{T_0} + \frac{2V \times 1.5P_0}{T_0} = \frac{P \times 5V}{T_0}$$

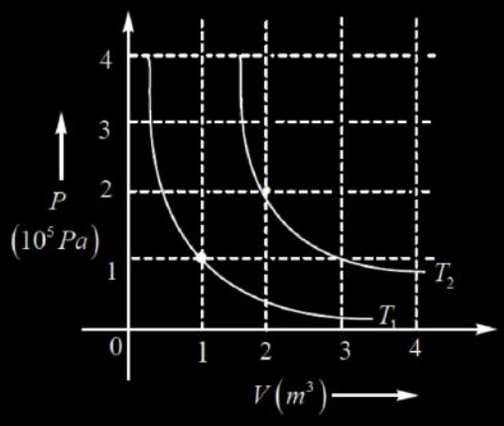
($\because \Delta W = 0$ & $\Delta Q = 0$ $\therefore \Delta T = 0$)

$$\Rightarrow P = 3P_0 = \frac{n}{2}P_0 \Rightarrow n = 6 \quad \text{Ans.}$$

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EXAMPLE:

The following graph shows two isotherms for a fixed mass of an ideal gas. The ratio of r.m.s speed of the molecules at temperatures T_1 and T_2 is _____ .

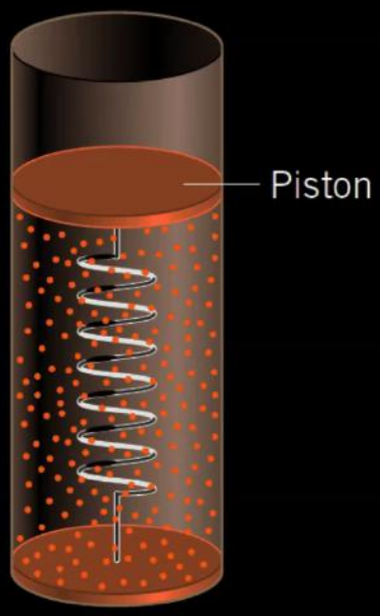


Sol: $v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{nM}} \propto \sqrt{PV}$

$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{P_1V_1}{P_2V_2}} = \sqrt{\frac{1\times1}{2\times2}} = \frac{1}{2}$ Ans.

EXAMPLE (CUTNELL):

26. M GO The drawing shows an ideal gas confined to a cylinder by a massless piston that is attached to an ideal spring. Outside the cylinder is a vacuum. The cross-sectional area of the piston is $A = 2.50 \times 10^{-3}$ m^2. The initial pressure, volume, and temperature of the gas are, respectively, P_0, $V_0 = 6.00 \times 10^{-4}$ m^3, and $T_0 = 273$ K, and the spring is initially stretched by an amount $x_0 = 0.0800$ m with respect to its unstrained length. The gas is heated, so that its final pressure, volume, and temperature are P_f, V_f, and T_f, and the spring is stretched by an amount $x_f = 0.1000$ m with respect to its unstrained length. What is the final temperature of the gas?



PROBLEM 26

Sol: $P_0 = kx_0/A$ $P_f = kx_f/A$ Now by mole conservation:

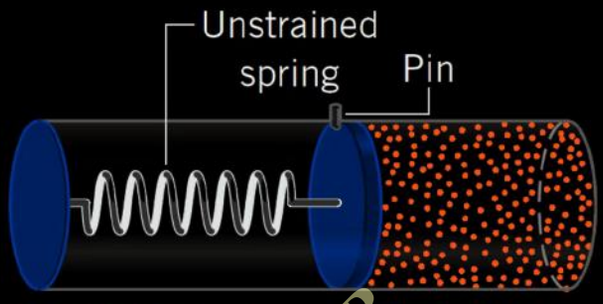
$\frac{P_iV_i}{T_i} = \frac{P_fV_f}{T_f} \Rightarrow \frac{kx_0 \times V_0}{A\ T_0} = \frac{kx_f\,(V_0 + A(x_f - x_i))}{A\ T_f}$

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$$\Rightarrow \quad T_f = \frac{T_0\, x_f\,(V_0 + A(x_f - x_{i0}))}{x_0\, V_0} \quad \text{Ans.}$$

EXAMPLE (Cutnell):

64. H A gas fills the right portion of a horizontal cylinder whose radius is 5.00 cm. The initial pressure of the gas is 1.01×10^5 Pa. A frictionless movable piston separates the gas from the left portion of the cylinder, which is evacuated and contains an ideal spring, as the drawing shows. The piston is initially held in place by a pin. The spring is initially unstrained, and the length of the gas-filled portion is 20.0 cm. When the pin is removed and the gas is allowed to expand, the length of the gas-filled chamber doubles. The initial and final temperatures are equal. Determine the spring constant of the spring.



PROBLEM 64

Sol: By mole conservation:

$$\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f} \Rightarrow P_f = P_i \frac{V_i}{V_f} \times \frac{T_f}{T_i}$$

$$\Rightarrow P_f = \frac{P_0}{2} = \frac{k x}{\pi r^2} \Rightarrow k = \frac{P_0 \pi r^2}{2x}, \quad x = 20\text{ cm}, \; P_0 = 1\text{ atm}, \; r = 5$$

EXAMPLE: (Pathfinder)



13. A highly conducting cylindrical vessel placed on elevated small supports contains an ideal gas under a piston of mass m. Area of the base of the cylinder is A. The piston is in equilibrium in the mid of the cylinder. Friction force between the cylinder and the piston can be ignored. Total mass of the cylinder and the gas is M. The atmospheric pressure is p_0. Now the piston is slowly pulled upwards, find the maximum value of M such that the cylinder can be lifted off the supports.

(a) $\frac{p_0 A - mg}{g}$ (b) $\frac{p_0 A - mg}{2g}$

(c) $\frac{2p_0 A - mg}{2g}$ (d) It is impossible in this way.

Solution: Here highly conducting vessel implies isothermal process. As the piston is pulled out, pressure decreases in the vessel and least pressure occurs when the piston is at mouth of cylinder.

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Initial pressure of gas $= p_0 + mg/A$

When at mouth of vessel, $p' = \dfrac{(p_0 + mg/A)}{2}$ ($\because PV = $ Constt & Volume doubles).

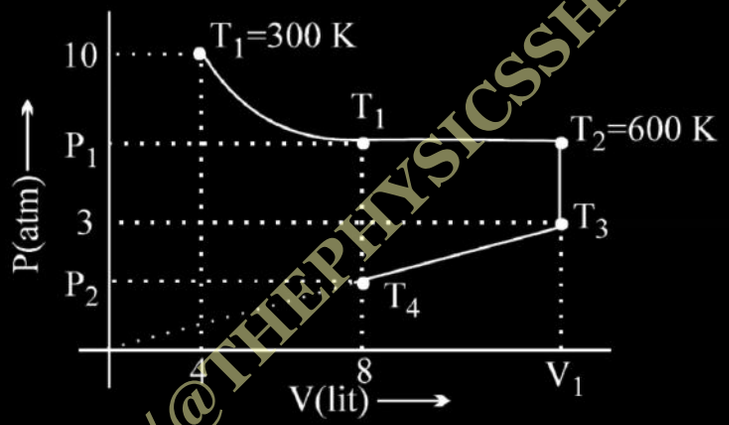
Now balancing forces on vessel:

$$p_0 \times A - \left(\frac{p_0 + mg/A}{2}\right) \times A = Mg.$$

$$\Rightarrow M = \frac{p_0 A - mg}{2g} \quad \text{Ans.} \quad (b).$$

EXAMPLE:

Fixed mass of a gas is subjected to the changes as shown is diagram, calculate T_3, T_4, P_1, P_2 and V_1 as shown is diagram. Considering gas obeys $PV = nRT$ equation.



Solution:

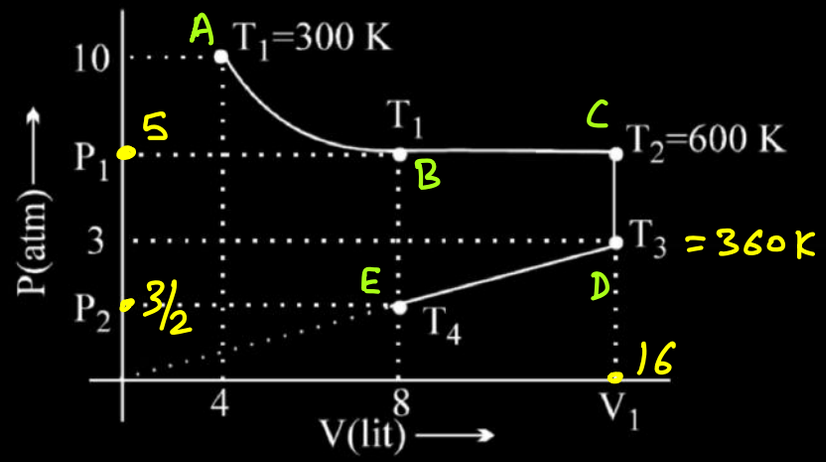
A→B Isothermal

$\Rightarrow PV = $ Constt

$\Rightarrow 10 \times 4 = P_1 \times 8 \Rightarrow P_1 = 5$

B→C Isobaric $\Rightarrow T/V = $ Constt

$\Rightarrow \dfrac{300}{8} = \dfrac{600}{V_1} \Rightarrow V_1 = 16$



Now C→D Isochoric $\Rightarrow P/T = $ Constt.

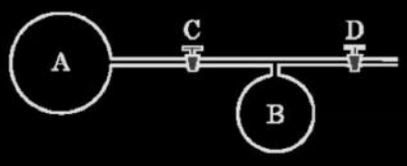
$\Rightarrow \dfrac{5}{600} = \dfrac{3}{T_3} \Rightarrow T_3 = 360\ K$

Now D→E : $P/V = $ Constt.

$\Rightarrow \dfrac{3}{16} = \dfrac{P_2}{8} \Rightarrow P_2 = \dfrac{3}{2}$

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EXAMPLE: (Pathfinder)



14. Two flasks A and B made of conducting materials are connected by a thin tube. Two valves C and D are installed on the tube as shown in the figure. Initially, both the flasks are evacuated and the valves are closed. Now the valve D is opened, an ideal gas is filled in the flask B at pressure p and then the valve D is closed. When the valve C is opened, the pressure in the flask B drops by amount Δp. If volume of the flask B is V_B, what is the volume of the flask A?

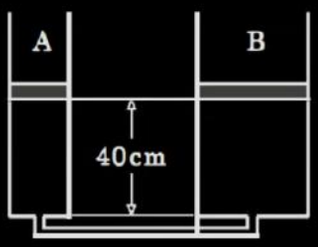
(a) $\frac{\Delta p V_B}{p-\Delta p}$ (b) $\frac{p V_B}{p-\Delta p}$

(c) $\frac{(p-\Delta p)V_B}{p}$ (d) $\frac{(p-\Delta p)V_B}{\Delta p}$

Solution: By mole conservation: $\frac{p V_B}{T} = \frac{(p-\Delta p)(V_A + V_B)}{T}$

$\Rightarrow V_A = \frac{p}{(p-\Delta p)} V_B - V_B = \left(\frac{\Delta p}{p-\Delta p}\right) V_B$ Ans (a)

EXAMPLE (pathfinder):



15. Two vertical conducting cylinders A and B of different cross sections are connected by a thin tube as shown. An ideal gas is trapped in the cylinders by two airtight pistons of masses 1.0 kg and 2.0 kg respectively. Initially the pistons are at the same height 40 cm above the base of the cylinders. If an additional mass of 10 g were gently placed on the piston in the cylinder A, which of the following statements correctly describe the new steady state ultimately reached.

(a) Pressure of the gas has been increased.
(b) Pressure of the gas remains unchanged.
(c) Height difference between the pistons becomes 30 cm.
(d) Height difference between the pistons becomes 60 cm.

Solution: $m_A = 1Kg$, $m_B = 2Kg$, For equilibrium: $\frac{m_A g}{A_A} = \frac{m_B g}{A_B}$

$\Rightarrow A_B = 2A_A$. Now when additional weight is placed on A, it must move down & B must move up and in steady state Gauge pressure of gas must be still $m_B g/A_B$ $\therefore$ pressure of the gas remains unchanged. (Piston A must fall all the way to bottom) Now Since P & T are same, V must be same. Let final height in B be H

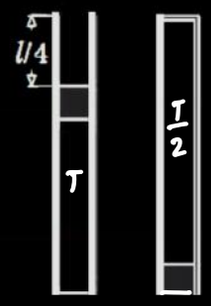
$\Rightarrow 2AH = A \times 40cm + 2A \times 40cm$

$\Rightarrow H = 60cm$ $\therefore$ (b) and (d).

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EXAMPLE: (PATHFINDER)

A tube of length l closed at lower end is held vertically. If an airtight piston is placed on its mouth, the piston slides down without friction and stops with its upper face at a depth $l/4$ from the mouth of the tube. If the temperature of air inside the tube is halved and the tube is inverted, the piston will settle with its lower face in level with the mouth of the tube as shown in the figure. Find thickness of the piston.



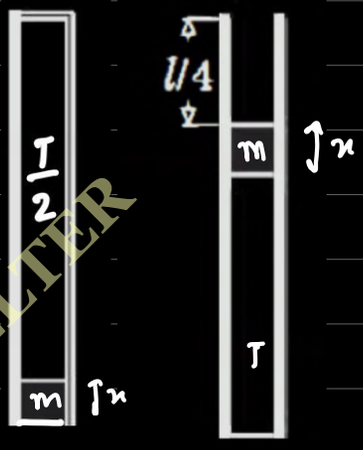
Solution: By mole conservation: (let $m_0/A = p_m$)

$$\frac{(p_0 + p_m)\,x(3l/4 - x)}{T} = \frac{(p_0 - p_m)(l - x)}{(T/2)} \quad (1)$$

($\because$ Volume $\propto$ length)

[Now when piston was put on mouth, until final settling position is also isothermal states) $\therefore$

Also: $p_0 \times l = (p_0 + p_m)(3l/4 - x)$ (2)



Solving (1) & (2): $x = (2 - \sqrt{3})\,l/4$ Ans

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